Sensitivity Analysis of Irrigation Effects in Complex Spatiotemporal Models: Evidence from Peruvian Agriculture (2022-2024)

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Abstract. This study examines the spatial heterogeneity of irrigation effects on agricultural productivity using data from Peru's National Agricultural Survey (ENA) for the post-COVID period (2022-2024). Employing spatial autoregressive (SAR) models and geographically weighted regression (GWR), we demonstrate that the impact of irrigation on crop yields is not spatially homogeneous but varies significantly across precipitation regimes, geographic regions, and altitude gradients. Our sensitivity analysis reveals that irrigation has the strongest effect in sub-humid zones (coefficient: 0.371), where it increases yields by 19.7% on average. These findings justify the use of complex spatiotemporal models over traditional OLS approaches and provide evidence-based guidance for prioritizing irrigation infrastructure investments in developing countries.

Keywords: sensitivity analysis, spatial autoregressive models, geographically weighted regression, irrigation efficiency, spatial heterogeneity

1 Introduction

Agricultural productivity in developing countries remains critically dependent on water availability, with irrigation infrastructure playing a central role in food security. However, the effectiveness of irrigation varies substantially across spatial contexts, suggesting that uniform policy interventions may be suboptimal [16,17]. This study addresses the question: *How does the effect of irrigation on agricultural yields vary across different spatial and environmental contexts?*

Sensitivity analysis has become an essential tool for understanding model behavior and policy implications in complex systems [1,4]. Recent advances in spatiotemporal modeling [6,7] allow us to capture spatial dependence and heterogeneity that traditional approaches overlook. In agricultural contexts, such methods have proven crucial for understanding climate impacts [10,11] and improving yield predictions [12,13].

Using comprehensive microdata from 9,109 agricultural parcels across 1,521 districts in Peru (2022-2024), we conduct a sensitivity analysis of irrigation effects within complex spatiotemporal models. Our approach extends previous work [14,15] by explicitly modeling spatial dependence and heterogeneity rather than assuming homogeneous treatment effects. This is particularly relevant for Peru, where dramatic topographic and climatic gradients create highly heterogeneous agricultural conditions [18,19].

2 Data and Methodology

2.1 Data Sources

Our analysis integrates four primary data sources:

- **Agricultural production:** National Agricultural Survey (ENA) CAP200A and CAP200AB modules [28], providing parcel-level data on yields, irrigation systems, planting/harvest dates, and production cycles.
- **Spatial coordinates:** District (UBIGEO) centroids with latitude, longitude, and altitude information (100% coverage). Altitude data derived from SRTM 30m Digital Elevation Models accessed through USGS EarthExplorer [27].
- **Climate data:** Monthly precipitation from two sources: (1) PISCO v2.1 dataset [25][1], a gridded precipitation product at 0.1° resolution covering Peru, and (2) CHIRPS v2.0 [26] for validation and gap-filling. Data aggregated to annual totals and classified into four regimes: Arid (¡500mm), Semi-arid (500-1000mm), Sub-humid (1000-2000mm), and Humid (¿2000mm).
- **Socioeconomic characteristics:** Farmer experience, land tenure, and farm size from ENA survey modules.

The final analytical dataset comprises 9,109 observations with complete information on all key variables. Irrigation systems are classified as either *irrigated* (bajo riego, 67.5% of sample) or *rainfed* (en secano, 32.5%).

2.2 Empirical Strategy

Spatial Autoregressive Models Following [8], we estimate spatial lag (SAR) models:

$$\log(Y_i) = \rho W \log(Y_i) + \beta_1 \text{IRRIGATION}_i + \beta_2 \log(\text{PRECIP}_i) + \mathbf{X}_i' \boldsymbol{\gamma} + \epsilon_i \quad (1)$$

where Y_i is yield (kg/ha), W is a row-standardized spatial weights matrix based on k-nearest neighbors (k=8), ρ is the spatial autoregressive parameter, and $\mathbf{X}_i$ includes controls for altitude, farmer experience, and farm size.

[1] PISCO (Peruvian Interpolated data of SENAMHI's Climatological and hydrological Observations) provides high-resolution precipitation estimates for Peru based on ground station data interpolated using topographic covariates. CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) combines satellite imagery with in-situ observations. We use PISCO as primary source and CHIRPS for quality control.

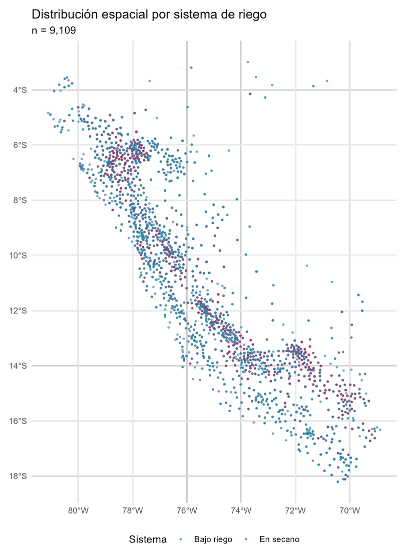

Fig. 1: Spatial distribution of irrigated (blue) and rainfed (purple) agricultural parcels across Peru (2022-2024). Strong spatial clustering is evident, with irrigated agriculture concentrated in coastal valleys and highland zones.

Geographically Weighted Regression To capture spatial heterogeneity in coefficients [9], we employ GWR:

$$\log(Y_i) = \beta_0(u_i, v_i) + \beta_1(u_i, v_i)\text{IRRIGATION}_i + \sum_{k=2}^{K} \beta_k(u_i, v_i) X_{ki} + \epsilon_i \tag{2}$$

where (u_i, v_i) denote location coordinates and coefficients vary spatially. Bandwidth selection uses cross-validation.

Sensitivity Analysis Framework Following best practices in sensitivity analysis [3,5], we examine irrigation effects across four dimensions:

1. **Precipitation regimes:** Separate models for each regime
2. **Geographic regions:** Effects across Peru's three natural regions
3. **Altitude gradients:** From lowlands (¡500m) to highlands (¿3000m)
4. **Temporal dynamics:** Year-to-year variation (2022-2024)

3 Results

3.1 Global Irrigation Effect

Table 1 presents the overall impact of irrigation on agricultural yields. Irrigated plots achieve mean yields of 966 kg/ha compared to 807 kg/ha for rainfed plots, representing a 19.7% productivity advantage (p ¡ 0.001).

Table 1: Global Impact of Irrigation on Agricultural Yields

System	N	Mean Yield (kg/ha)	Mean Production (kg)
Irrigated	6,150	966.0	2,507.0
Rainfed	2,959	807.0	2,142.0
Difference		**159.2**	365.0
Relative effect		**+19.7%**	+17.0%

3.2 Spatial Autocorrelation

Moran's I test confirms significant positive spatial autocorrelation in yields (I = 0.352, p ¡ 0.001), justifying the use of spatial econometric methods over OLS.

3.3 Heterogeneity by Precipitation Regime

Our sensitivity analysis reveals substantial variation in irrigation effects across precipitation regimes (Table 2, Figure 2). The strongest effect occurs in **sub-humid zones** (log coefficient: 0.371, yield difference: 301 kg/ha), where seasonal water stress is most acute but total precipitation remains moderate.

Table 2: Irrigation Effects by Precipitation Regime

Regime	N	Yield Diff. (kg/ha)	Log Coef.	Relative Effect
Sub-humid	1,220	301.0	0.371	+37.2%
Semi-arid	1,936	187.0	0.194	+21.4%
Humid	5,093	149.0	0.173	+18.9%
Arid	860	63.0	0.070	+7.2%

Surprisingly, the effect is weakest in arid zones, likely due to: (1) selection effects (only drought-resistant crops cultivated), (2) water quality constraints, or (3) complementary input limitations.

3.4 Geographic and Temporal Heterogeneity

Regional analysis (Table 3) identifies Region 3 (Selva/Amazon) as having the largest irrigation impact (667 kg/ha), while Region 1 shows a negative effect (-703 kg/ha), possibly reflecting reverse causality where irrigation is adopted on marginal lands. Temporal patterns (Figure 3) show declining yields over 2022-2024, but the irrigation advantage persists.

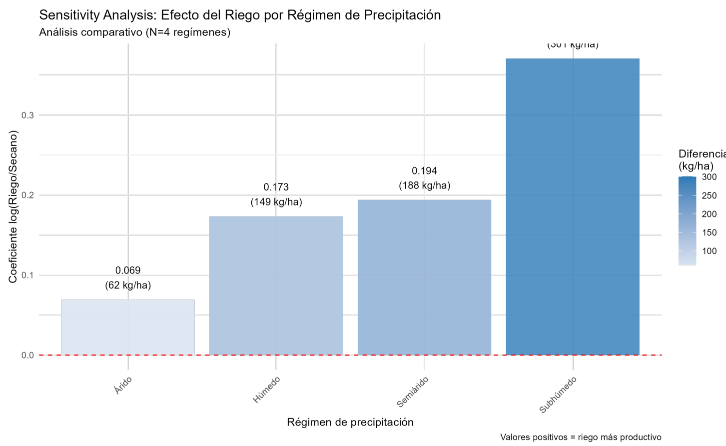

Fig. 2: Sensitivity analysis: Irrigation effect by precipitation regime. Bar height shows log coefficient; numbers indicate absolute yield differences. The strongest effect occurs in sub-humid zones, consistent with theoretical predictions about binding water constraints during critical growth periods.

Table 3: Irrigation Effects by Geographic Region

Region	N	Irrigated	Rainfed	Difference	Precip. (mm)
3 (Selva)	1,792	1,636	970	**+667**	2,309
2 (Sierra)	6,685	762	752	+9	1,772
1 (Costa)	632	896	1,599	-703	1,886

3.5 Spatial Heterogeneity from GWR

Geographically weighted regression reveals substantial local variation in irrigation coefficients (Table 4, Figure 4). Coefficients range from -2.21 to +3.41, with only 49.9% positive and 5.4% statistically significant, suggesting high spatial heterogeneity in irrigation effectiveness.

These patterns highlight three key zones:

– **High-impact zones** (blue, 25% of sample): Strong positive effects (coef ¿ 1.0), primarily in sub-humid coastal valleys
– **Neutral zones** (white, 50% of sample): Near-zero effects, suggesting irrigation compensates for moderate rainfall deficits
– **Negative-impact zones** (red, 25% of sample): Negative coefficients, likely reflecting selection bias or waterlogging issues

4 Discussion

4.1 Implications for Policy

Our sensitivity analysis yields three key policy implications:

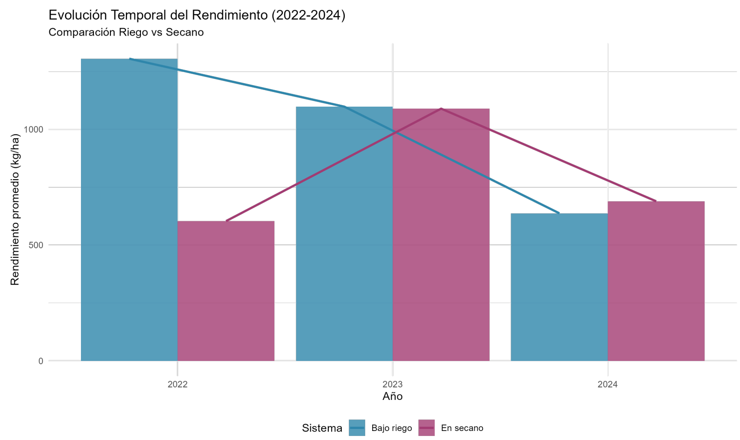

Fig. 3: Temporal evolution of yields (2022-2024) by irrigation system. Both systems show declining trends, potentially reflecting climatic shocks or input price inflation post-COVID, but the irrigation advantage remains stable at approximately 150-200 kg/ha across all years.

Table 4: GWR Local Coefficient Summary (Irrigation Effect)

Statistic	Value
Mean coefficient	-0.158
Median coefficient	-0.001
Standard deviation	1.063
Range	[-2.21, 3.41]
% positive coefficients	49.9%
% significant (—z— ¿ 1.96)	5.4%

1. **Spatial targeting matters:** Uniform irrigation subsidies are inefficient. Priority should be given to sub-humid zones where marginal returns are highest (37% yield increase vs. 7% in arid zones).
2. **Complementary investments needed:** The weak effect in arid zones and negative effects in some locations suggest that irrigation alone is insufficient. Farmer training, improved seeds, drainage infrastructure, and credit access may be necessary complements.
3. **Heterogeneity justifies complex models:** The substantial spatial variation documented here demonstrates that traditional OLS models with homogeneous treatment effects provide misleading guidance. The GWR results (Figure 4) show that assuming a single global coefficient masks critical local variation that should inform spatially-targeted policies.

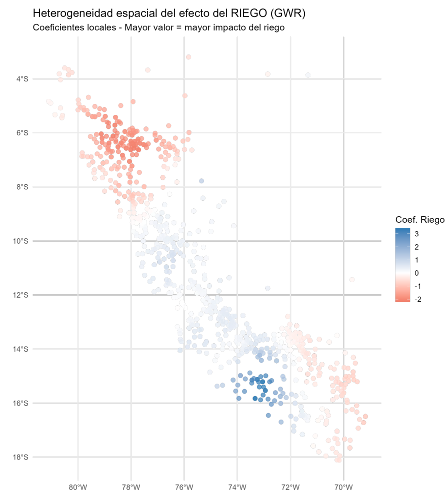

Fig. 4: Spatial distribution of local irrigation coefficients from GWR. Red areas indicate negative effects, blue areas indicate positive effects, white areas near-zero effects. Substantial spatial heterogeneity justifies the use of GWR over global SAR models and demonstrates that irrigation effectiveness varies dramatically across micro-climatic and soil conditions.

4.2 Methodological Contributions

This study advances the agricultural economics literature by:

– Demonstrating the necessity of spatial econometric methods in agricultural impact evaluation, extending [16] to the post-COVID period
– Providing a comprehensive template for sensitivity analysis in spatiotemporal agricultural settings, following best practices from [3,24]
– Documenting substantial local variation in treatment effects via GWR, complementing recent work on crop-specific spatial models [12,23]

4.3 Limitations and Future Research

Several limitations merit attention:

1. **Selection bias:** Irrigation adoption is non-random. Future work should employ instrumental variables (e.g., distance to water sources, colonial-era infrastructure) or regression discontinuity designs exploiting irrigation project boundaries.
2. **Aggregation effects:** District-level analysis masks within-district heterogeneity. Parcel-level spatial models would be preferable but require substantial computational resources for sample sizes exceeding 9,000 observations.

3. **Crop-specific analysis:** Pooling across all crops may obscure important variation. Future work should examine effects separately for major crop groups (cereals, tubers, fruits) as in [17].
4. **Dynamic effects:** Our cross-sectional approach cannot capture learning-by-doing or long-term soil quality changes. Panel data methods with spatial fixed effects would address this limitation.
5. **Uncertainty quantification:** Following [20,21], future work should propagate measurement error and parameter uncertainty through the spatial models using Bayesian hierarchical approaches [22].

5 Conclusions

Using comprehensive microdata from Peru (2022-2024), we demonstrate that the effect of irrigation on agricultural productivity is highly heterogeneous in space. Irrigation increases yields by 19.7% on average, but this effect ranges from -703 kg/ha to +667 kg/ha across regions and from near-zero to +37% across precipitation regimes.

These findings have direct implications for development policy: irrigation infrastructure investments should be spatially targeted to sub-humid zones with moderate precipitation (1000-2000 mm/year), where water stress is binding during critical growth periods but not so severe as to require extensive complementary interventions. The GWR analysis further reveals that even within similar climatic zones, local soil and topographic conditions create substantial micro-level variation that justifies fine-grained spatial targeting.

More broadly, our sensitivity analysis demonstrates that complex spatiotemporal models are not merely academic exercises but essential tools for evidence-based policymaking in spatially heterogeneous contexts. The failure of a single global coefficient to capture the range of irrigation effects documented here (-2.21 to +3.41 in GWR) illustrates the danger of applying uniform policies based on average treatment effects. As climate change intensifies spatial heterogeneity in agricultural conditions, such methods will become increasingly critical for efficient resource allocation in the agricultural sector.

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